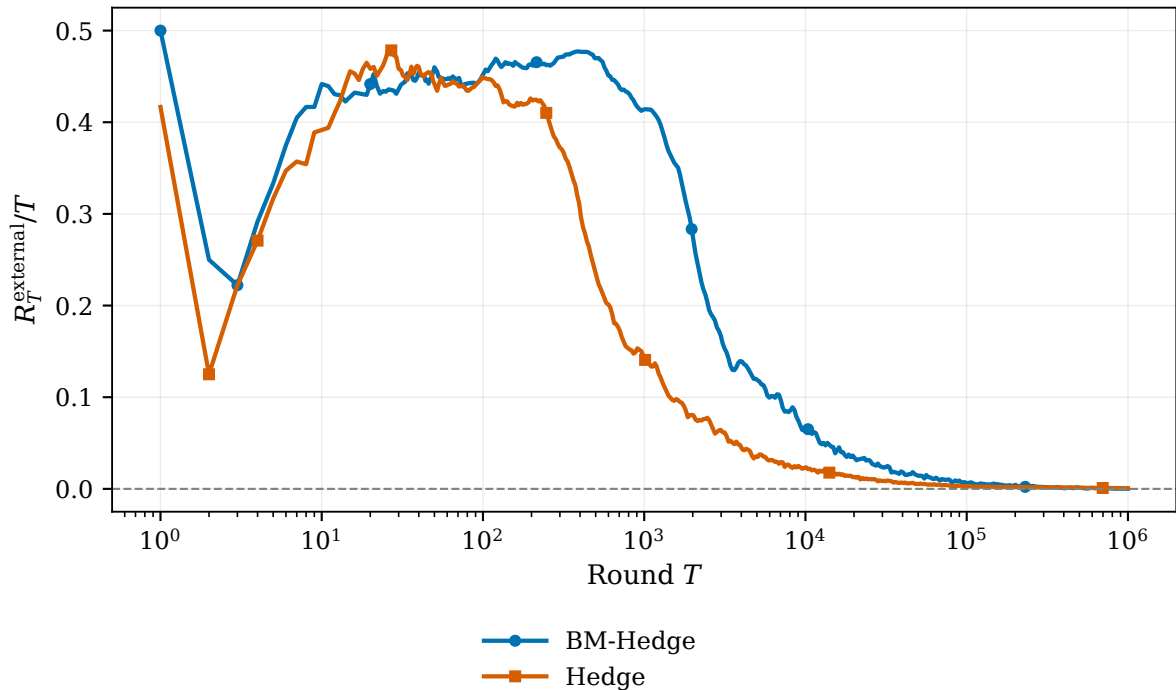


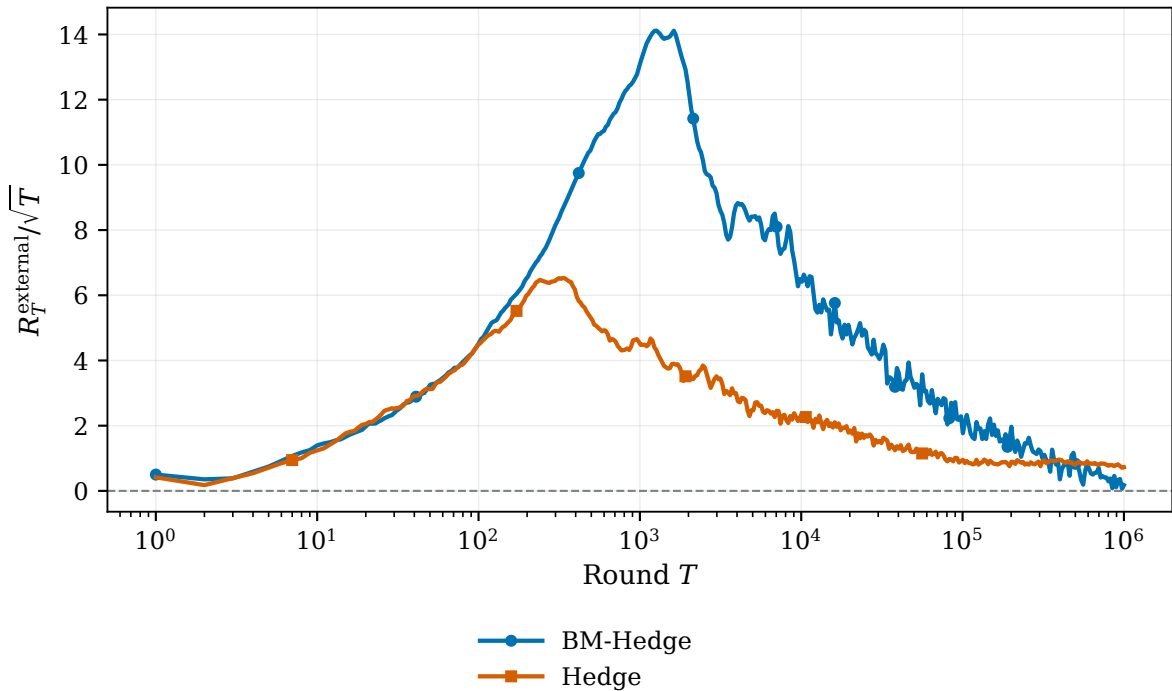
Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Full information
Algorithms:	BM-Hedge, Hedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	External regret
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge:	mean = 0.000175083, SD = 0.000465202, SE = 0.000134292
Hedge:	mean = 0.0007245, SD = 0.000326643, SE = 9.42938e-05



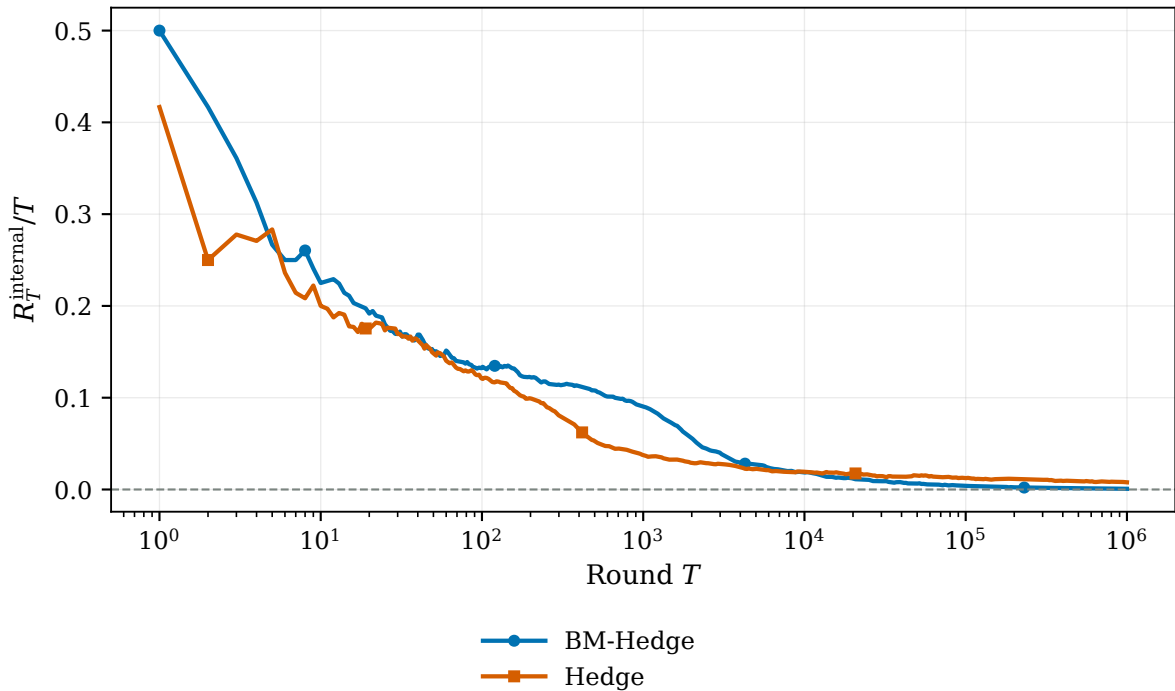
Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Full information
Algorithms:	BM-Hedge, Hedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	External regret
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge:	mean = 0.175083, SD = 0.465202, SE = 0.134292
Hedge:	mean = 0.7245, SD = 0.326643, SE = 0.0942938



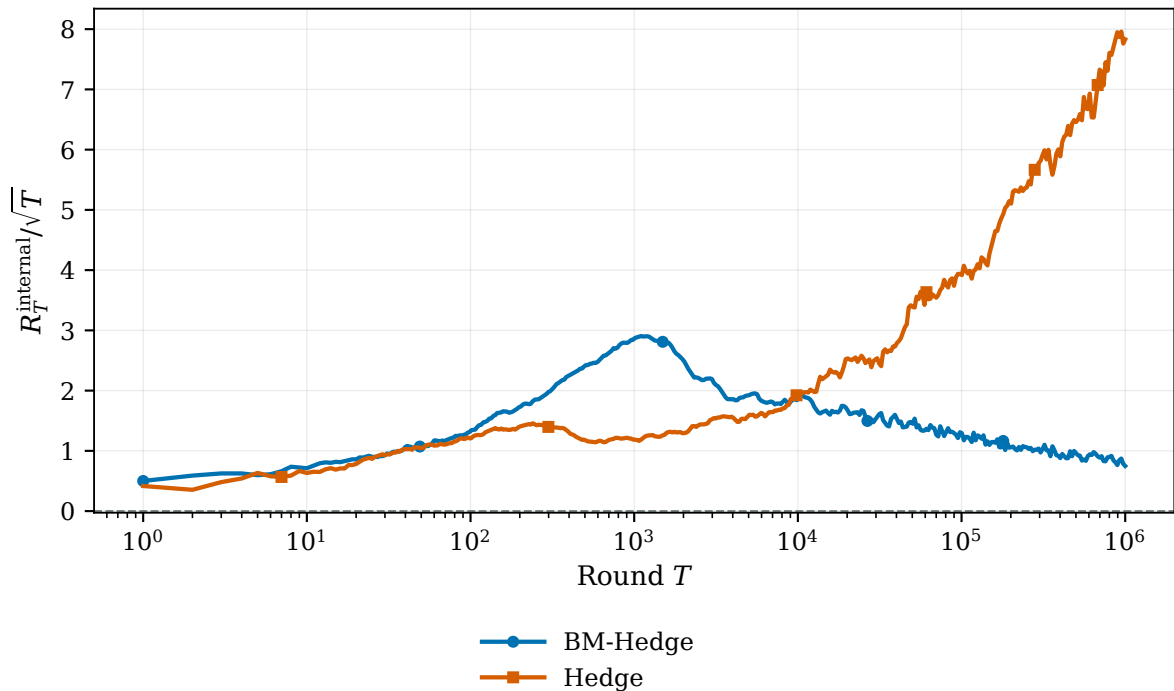
Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Full information
Algorithms:	BM-Hedge, Hedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Internal regret
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge:	mean = 0.000749333, SD = 0.000111192, SE = 3.20984e-05
Hedge:	mean = 0.00783042, SD = 0.00149371, SE = 0.000431196



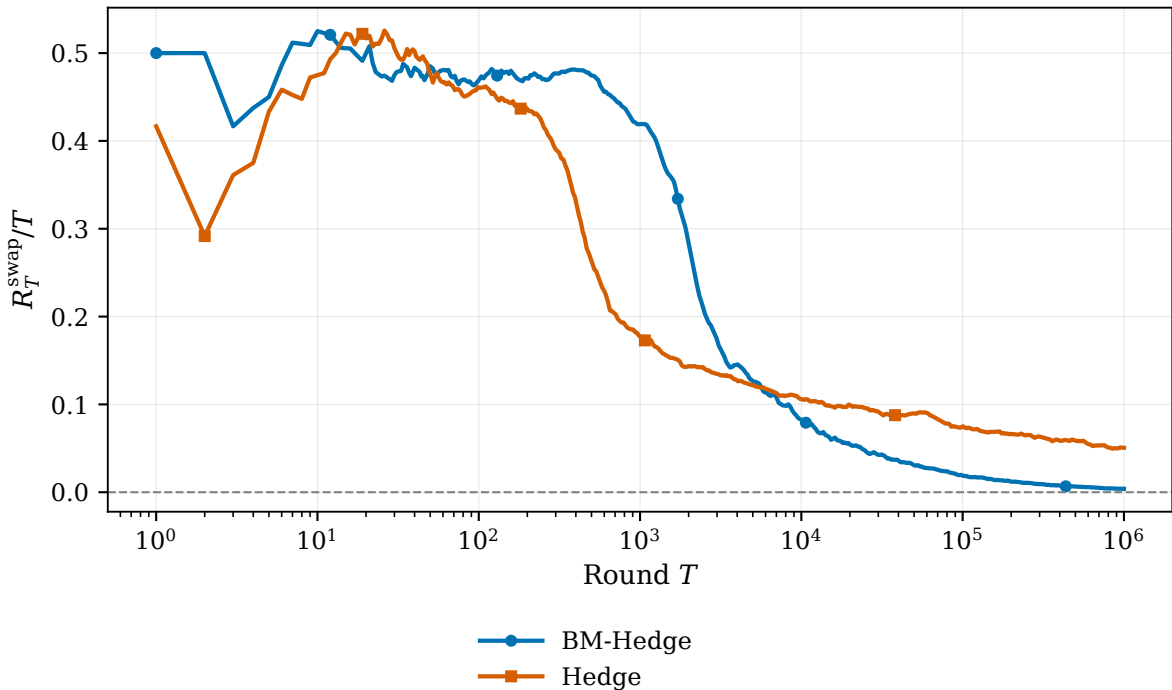
Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Full information
Algorithms:	BM-Hedge, Hedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Internal regret
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge:	mean = 0.749333, SD = 0.111192, SE = 0.0320984
Hedge:	mean = 7.83042, SD = 1.49371, SE = 0.431196



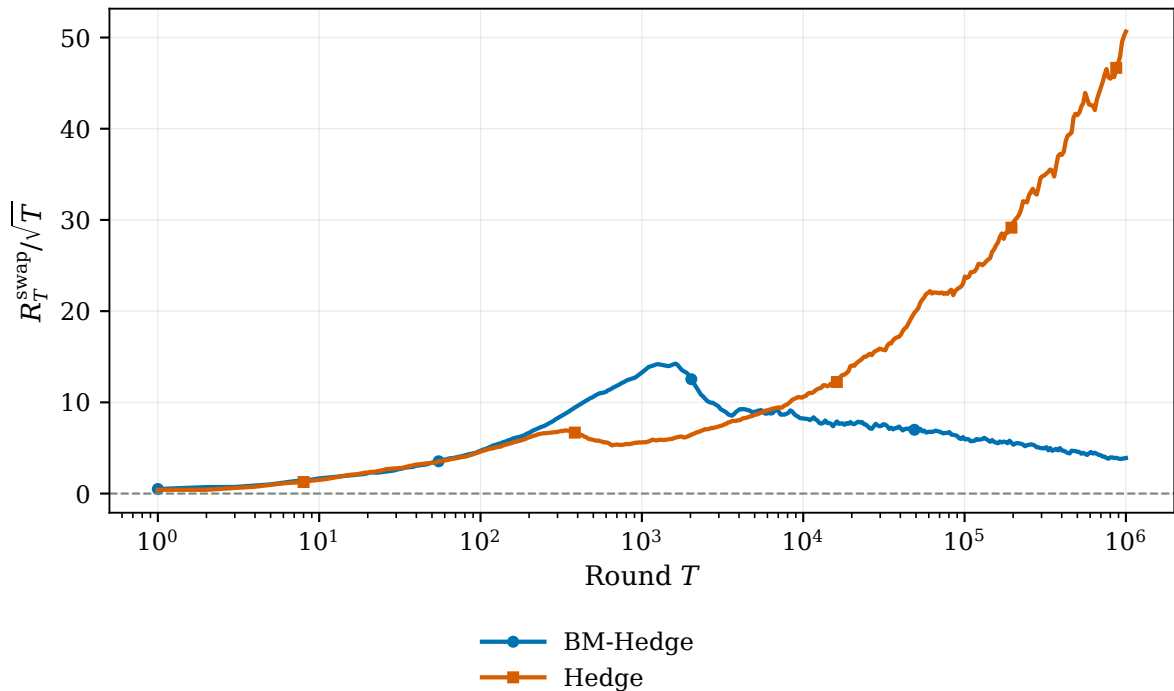
Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Full information
Algorithms:	BM-Hedge, Hedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Swap regret
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge:	mean = 0.00389417, SD = 0.000468079, SE = 0.000135123
Hedge:	mean = 0.0506502, SD = 0.00808444, SE = 0.00233378



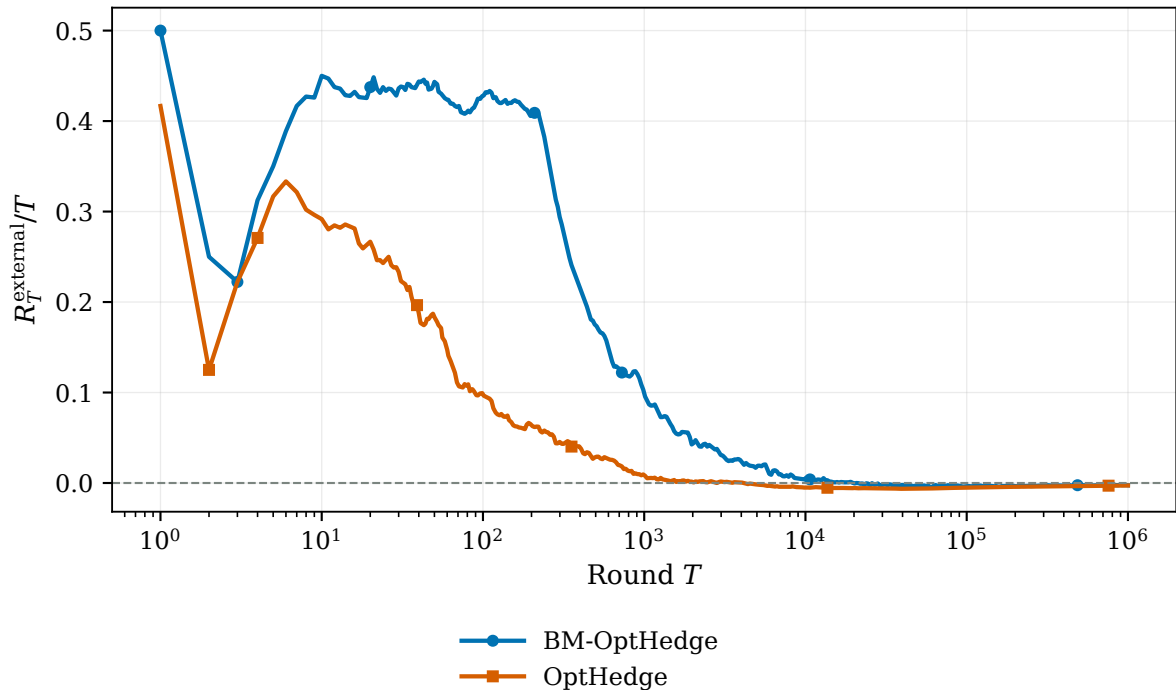
Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Full information
Algorithms:	BM-Hedge, Hedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Swap regret
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-Hedge:	mean = 3.89417, SD = 0.468079, SE = 0.135123
Hedge:	mean = 50.6502, SD = 8.08444, SE = 2.33378



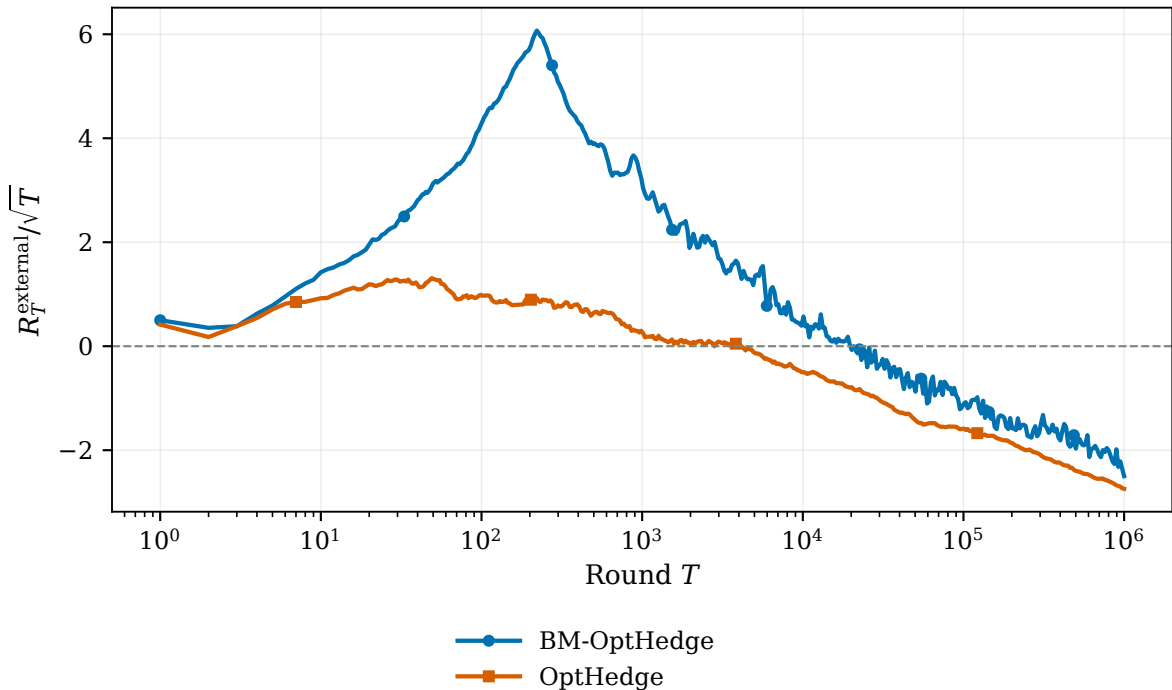
Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Full information
Algorithms:	BM-OptHedge, OptHedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	External regret
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-OptHedge:	mean = -0.00249917, SD = 0.00047666, SE = 0.0001376
OptHedge:	mean = -0.00274242, SD = 0.00049422, SE = 0.000142669



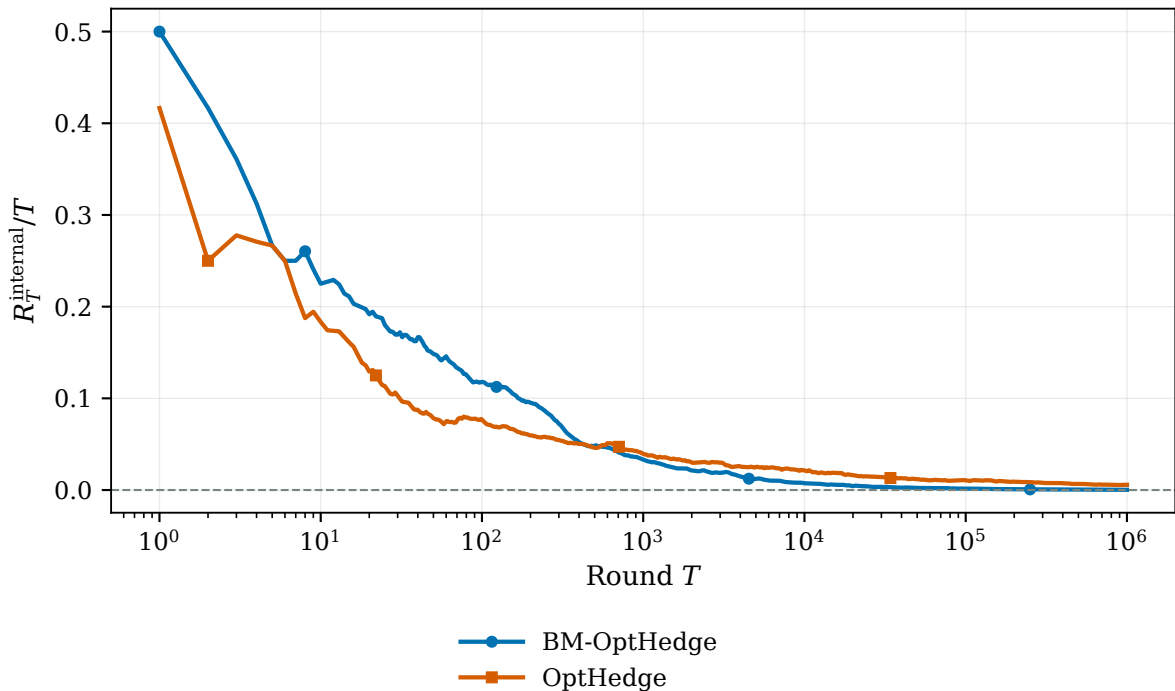
Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Full information
Algorithms:	BM-OptHedge, OptHedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	External regret
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-OptHedge:	mean = -2.49917, SD = 0.47666, SE = 0.1376
OptHedge:	mean = -2.74242, SD = 0.49422, SE = 0.142669



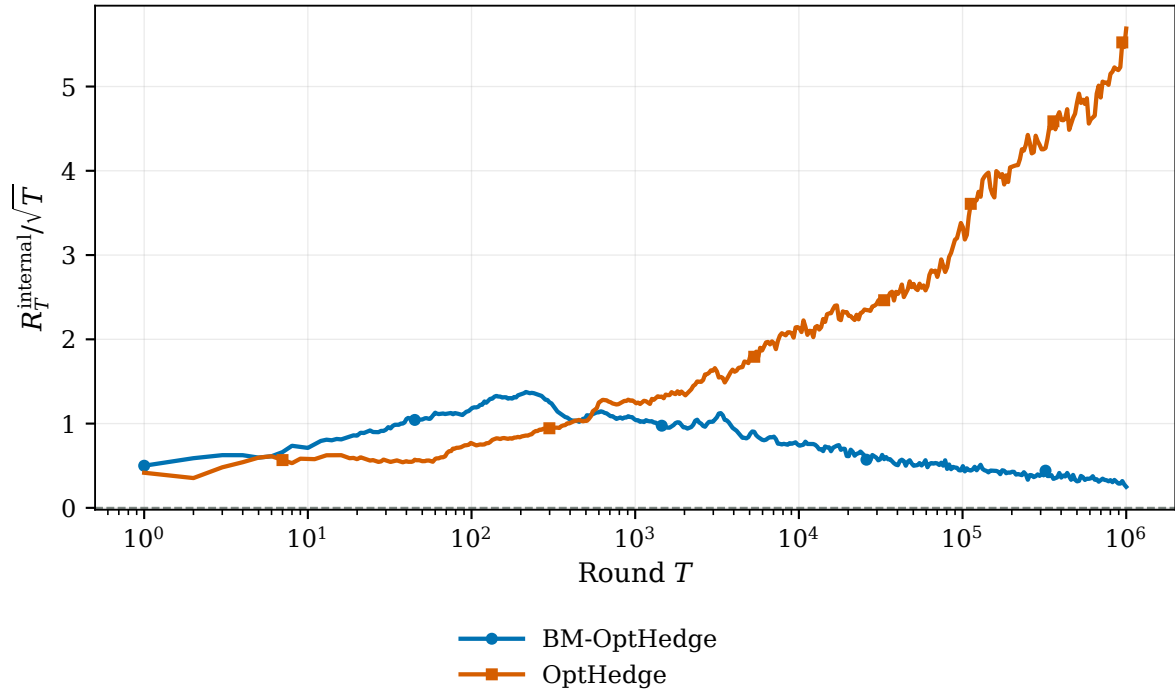
Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Full information
Algorithms:	BM-OptHedge, OptHedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Internal regret
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-OptHedge:	mean = 0.000249833, SD = 8.82711e-05, SE = 2.54817e-05
OptHedge:	mean = 0.00568767, SD = 0.000769671, SE = 0.000222185



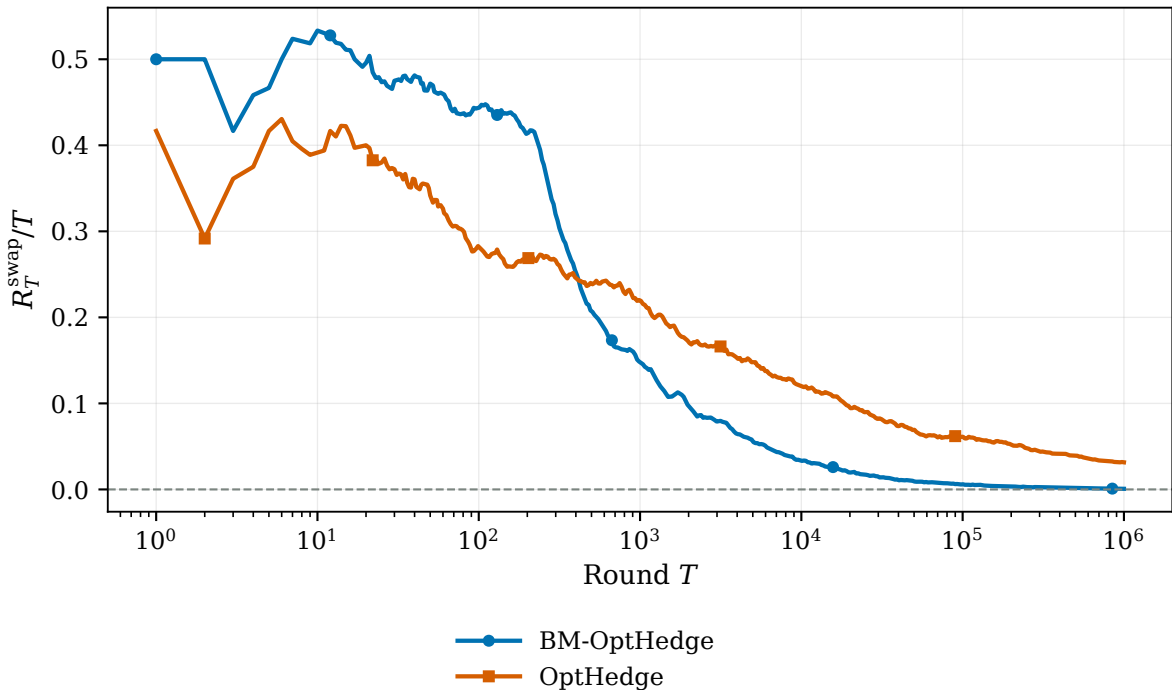
Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Full information
Algorithms:	BM-OptHedge, OptHedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Internal regret
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-OptHedge:	mean = 0.249833, SD = 0.0882711, SE = 0.0254817
OptHedge:	mean = 5.68767, SD = 0.769671, SE = 0.222185



Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Full information
Algorithms:	BM-OptHedge, OptHedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Swap regret
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-OptHedge:	mean = 0.000729333, SD = 0.000297851, SE = 8.59821e-05
OptHedge:	mean = 0.0315117, SD = 0.00529378, SE = 0.00152818



Experiment Information

Figure:	Algorithm comparison
Environment:	Historical-frequency
Actions:	10
Feedback:	Full information
Algorithms:	BM-OptHedge, OptHedge
Horizon:	1,000,000
Replicates:	12
Seed:	42
Metric:	Swap regret
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
BM-OptHedge:	mean = 0.729333, SD = 0.297851, SE = 0.0859821
OptHedge:	mean = 31.5117, SD = 5.29378, SE = 1.52818

